

JOSEPH PORTAS

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Jan 2026 - Aug 2026

M.S. Computer Science

Northeastern University, Khoury College of Computer Sciences

Sep 2023 - Dec 2025

B.S. Computer Science, Minor in Mathematics, Cum Laude

GPA: 3.6 | Dean's List: Fall 2023, Fall 2024, Spring 2025

TECHNICAL SKILLS

Programming Languages: Python | TypeScript | JavaScript | Java | SQL | C++ | C | Bash | HTML | CSS

Frameworks & Tools: Git | Docker | AWS | React | Node.js | FastAPI | Express | GitHub Actions | PostgreSQL | Pytest

EXPERIENCE

[Software Engineering Consultant - Northeastern Registrar](#), Boston, MA

Sep 2025 - Jan 2026

- Collaborated with Northeastern Registrar stakeholders, the Vice Provost, and a 7-person Agile team to build a full-stack exam scheduling platform replacing a paper-based registrar workflow, deployed with Docker, GitHub Actions, AWS ECS Fargate, RDS, and S3
- Built workflows with Next.js, TypeScript, FastAPI, and PostgreSQL to automate timetable creation for 15,000+ students, 1,500+ course sections, and 270 classrooms, while enforcing department, room, and time constraints
- Implemented a DSATUR graph-coloring scheduling engine to generate conflict-free schedules in minutes, preventing enrollment conflicts

Software Engineer Intern - BAE Systems, Endicott, NY

May 2025 - Jan 2026

- Developed a modular PySide6 GUI and backend for a hardware testing system, projected to save \$750K+ annually and reduce test setup time through reusable tooling
- Refactored internal Python scripts into reusable modules to improve extensibility and onboarding
- Built tools and automation scripts for firmware engineers, eliminating repetitive dev-testing tasks

Freelance Tutor - Mignano Consulting, Remote

Dec 2024 - Present

- Provide individualized instruction in SAT prep, Regents exams, and college-level STEM courses
- Help expand Mignano Consulting by designing tutoring classes, tracking student progress, and supporting recurring clients

PROJECTS

[Functional Core Compiler](#), Duo Project

Mar 2025 - Mar 2025

- Implemented a compiler for an extended functional programming language in Racket, translating parsed AST expressions into executable closures with support for lexical scoping, first-class functions, conditionals, recursion, mutation, and multi-expression function bodies
- Optimized compiled execution by resolving variable bindings at compile time using de Bruijn-style environment indexes and global binding inlining, reducing repeated runtime environment traversal and identifier lookup
- Extended the language runtime with mutable variable storage, by-reference function calls, recursive definitions, error handling, and semantic tests on closures, mutation, recursion, shadowing, and invalid programs

File System, Duo Project

Nov 2024 - Dec 2024

- Built a FUSE-based file system in C supporting create, read, write, delete, rename, mkdir, and rmdir operations
- Implemented block allocation, inode metadata, directory traversal, and disk-space management to support reliable storage up to 500KB per file
- Tested system behavior, command-line operations, and edge cases, improving correctness across path handling, file updates, and storage limits

LEADERSHIP

Teaching Assistant - Northeastern University, Boston, MA

Sep 2024 - May 2025

- Led labs and office hours and graded coursework for 100+ students in Fundamentals of Computer Science I
- Taught functional programming and software design principles

Resident Assistant - Northeastern University, Boston, MA

Aug 2025 - Present

- Support 30+ residents through community programming, event coordination, student support, and conflict resolution